EXPERIMENT 9: Word Embedding Visualization using PCA, t-SNE, and UMAP

AIM:

To generate random word embeddings for a sample corpus and visualize them using three dimensionality reduction techniques: Principal Component Analysis (PCA), t-Distributed Stochastic Neighbor Embedding (t-SNE), and UMAP.

PROCEDURE:

1. Import numpy, re, PCA from sklearn, TSNE from sklearn, matplotlib.
2. Define sample text and preprocess it.
3. Build vocabulary and create word2idx mapping.
4. Initialize random embeddings with dimension 5 for each word.
5. PCA: Apply PCA(n_components=2) on the embeddings.
6. t-SNE: Apply TSNE(n_components=2, perplexity=3) on embeddings.
7. UMAP (optional): Apply umap.UMAP(n_components=2) if available.
8. Define plot(data, title): scatter plot each word with its label.
9. Plot PCA and t-SNE results; try UMAP, print install message if unavailable.

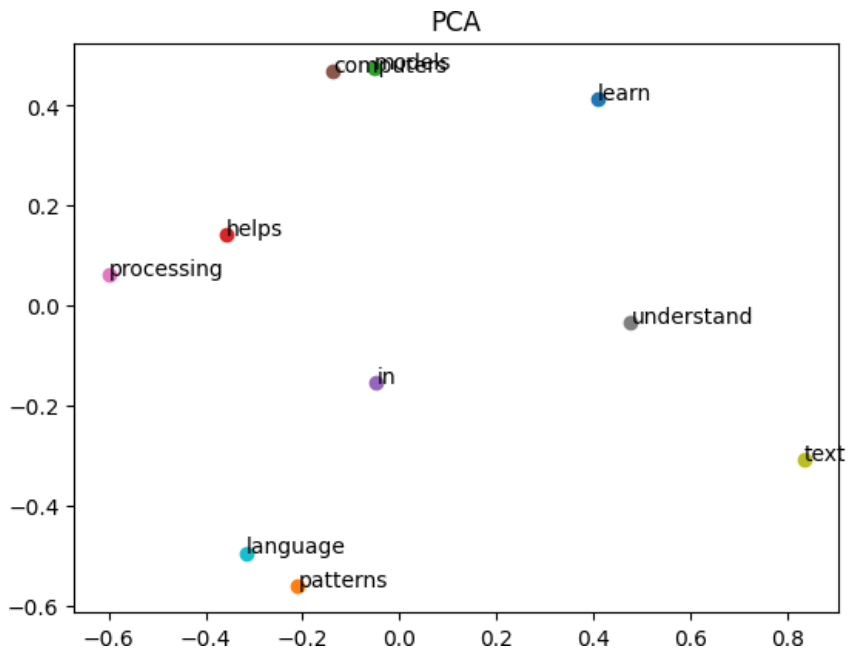
CODE:

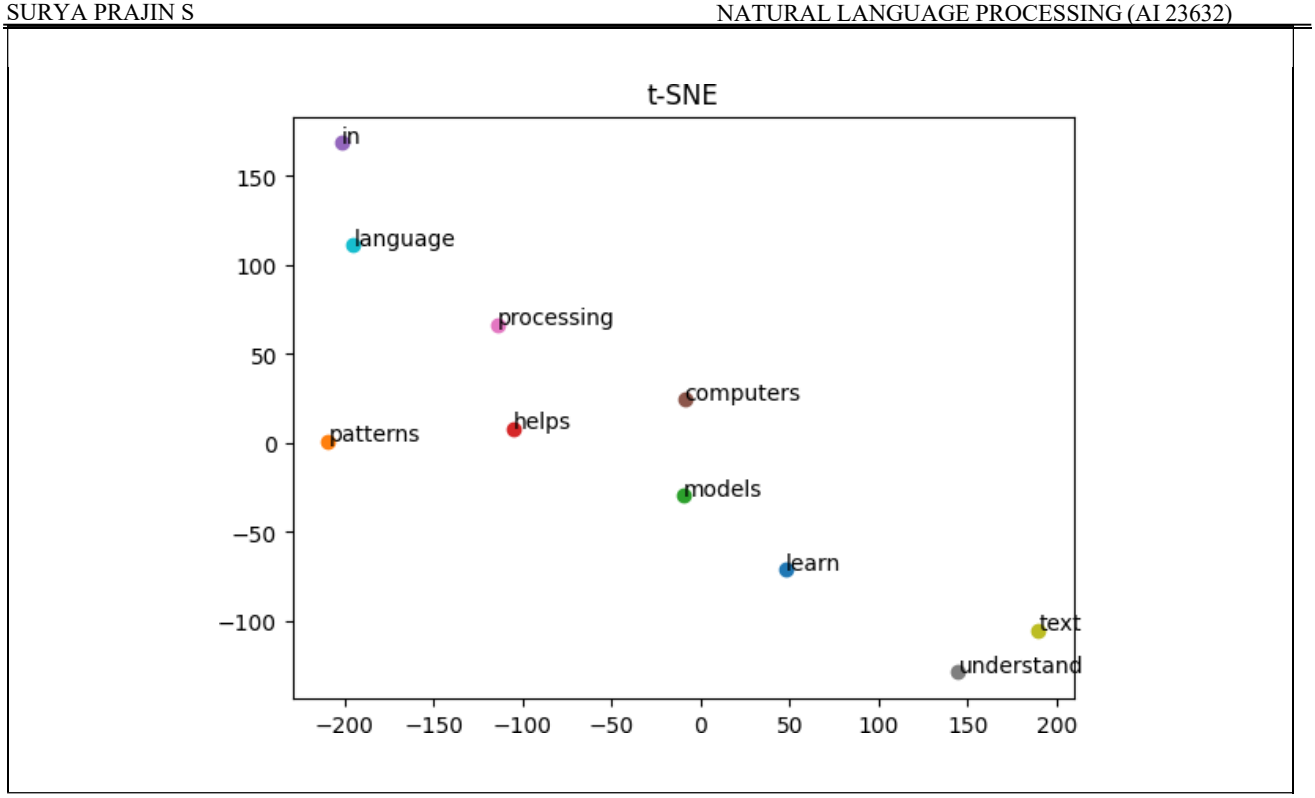
```
import numpy as np
import re
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt

text = "language models learn patterns in text language processing helps computers
understand language"

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    return text.split()

tokens = preprocess(text)

vocab = list(set(tokens))
word2idx = {w:i for i,w in enumerate(vocab)}

embedding_dim = 5
embeddings = np.random.rand(len(vocab), embedding_dim)

# PCA
pca = PCA(n_components=2)
pca_result = pca.fit_transform(embeddings)
```

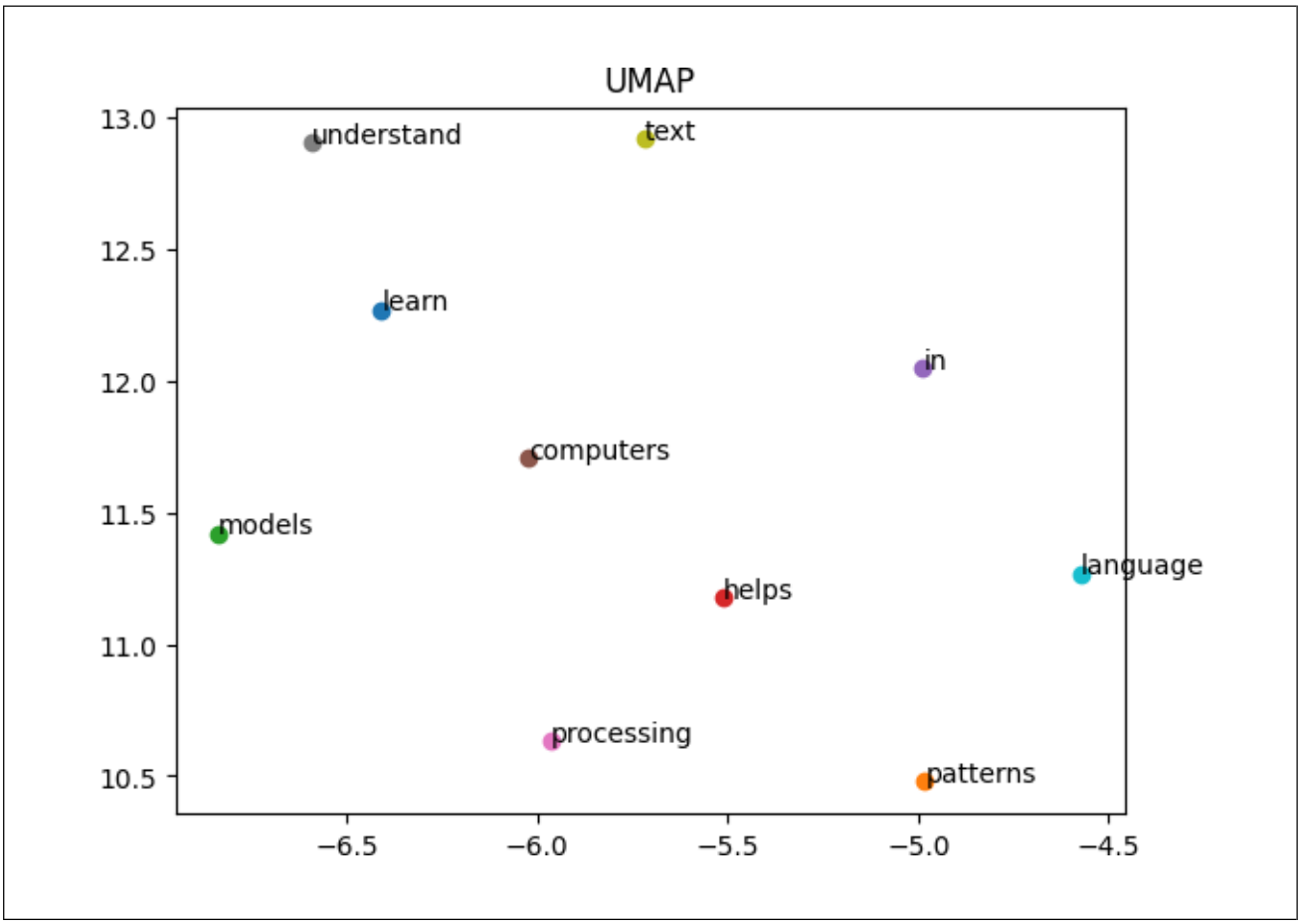
```
# TSNE
tsne = TSNE(n_components=2, perplexity=3)
tsne_result = tsne.fit_transform(embeddings)

def plot(data, title):
    plt.figure()
    for i, word in enumerate(vocab):
        plt.scatter(data[i,0], data[i,1])
        plt.text(data[i,0], data[i,1], word)
    plt.title(title)
    plt.show()

plot(pca_result, "PCA")
plot(tsne_result, "t-SNE")

try:
    import umap
    reducer = umap.UMAP(n_components=2)
    umap_result = reducer.fit_transform(embeddings)
    plot(umap_result, "UMAP")
except:
    print("Install UMAP: pip install umap-learn")
```

OUTPUT:

[PCA, t-SNE, and UMAP PLOTS - See Image Placeholder Below]




t-SNE
in
language
processing
computers
helps
patterns
models
learn
text
understand
150
100
50
0
−50
−100
−200
−150
−100
−50
0
50
100
150
200


UMAP
understand
text
learn
in
computers
models
language
helps
processing
patterns
13.0
12.5
12.0
11.5
11.0
10.5
−6.5
−6.0
−5.5
−5.0
−4.5

RESULT:

Word2Vec embeddings were successfully generated.Visualization techniques clearly showed word relationships.